Target-Conditioned Corpus Transformation: A Non-Neural Typological Projection

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Abstract

This paper documents a frozen French → Latin production run in Project RBT, a non-neural corpus transformation system that operates on corpus-level statistical fingerprints rather than cognate lists, sound laws, or explicit morphological supervision. The active system is target-conditioned: modern French is the source corpus and Latin provides the in-loop structural and form reference. Languages are represented through token co-occurrence, positional distributions, word n-gram profiles, and character-level form profiles. The reported *v5* run used 16 candidates per proposal, block size 1000, incremental scoring, and Fortran-accelerated hot-loop components. It halted natively by a plateau rule after 614,000 proposals. The final Latin structural score was −0.00010808926278147055 (bounded above by zero, with zero indicating an exact match), the final Latin form score was 0.7611594796180725, and the final family alignment score was 0.513866759690557 (approximately 54% of the achievable Latin ceiling under the active scoring function; see Limitations). The best form checkpoint occurred earlier in the run at block 338 with a Latin form score of 0.8632466197013855. Post hoc comparison against a held-out validator bank of six attested medieval Romance corpora showed a structural nearest-neighbor path that oscillated between Middle French and Old Spanish during the first half of the run before settling on Old Occitan in the second half, and a form nearest-neighbor path that alternated between Middle French and other langue d'oïl varieties. The paper reports the computational setup, stopping rule, preserved checkpoints, and validator-bank outputs for this run. A post hoc dimensional robustness check additionally places Sumerian, a non-Indo-European language unrelated to Romance, at fourth-rank structural proximity to Latin; we read the structural metric as a typological proximity detector rather than a chronological one.

1 Introduction

Project RBT studies target-conditioned statistical retrodiction in historical linguistic space. In the present configuration, a modern descendant corpus is iteratively mutated under explicit pressure from a historical target corpus, and the resulting synthetic states are compared against held-out attested historical corpora after the run completes. Most quantitative approaches to historical linguistic reconstruction operate on discrete lexical units: Bayesian phylogenetic methods using cognate sets and sound correspondences [1, 2], cognate-detection algorithms and standardized lexical databases for comparative work [4], statistical models of language change at the lexeme level [6], and computational-phylogenetic surveys that chart the methodological landscape [3]. The system reported here is different in kind: it operates on corpus-level statistical fingerprints (n-gram coverage, type-token statistics, character profiles, and suffix profiles) and makes no use of cognate alignments, sound laws, or hand-curated correspondences. The empirical interest of the trajectory lies in what intermediate states the fingerprint geometry produces under target pressure, not in what historical processes those states represent.

Many quantitative approaches to historical linguistic comparison and reconstruction now incorporate neural components, from Bayesian phylogenetic methods with neural posteriors to transformer-based cognate detection. The approach reported here is deliberately non-neural for three reasons. First, every feature in the structural and form scoring paths has an explicit statistical interpretation that can be read directly from a corpus and traced back from any score; this transparency is harder to achieve with learned embeddings. Second, the engine runs on CPU, is fully deterministic given a seed, and has no GPU, framework-version, or pre-trained-model dependencies; the v5 run reported below completed in approximately 135 hours of CPU compute and can be reproduced on commodity hardware. Third, the research question being investigated, what intermediate states a corpus-level fingerprint geometry produces under target pressure, is a property of the metric space rather than a predictive task; neural methods are well-suited to optimizing predictive accuracy on held-out test sets, but the question explored here is different in kind.

The present paper is descriptive in scope. It documents one frozen production run from modern French toward Latin and reports three classes of output:

1. the computational and statistical method used for the run,
2. the endpoint and checkpoint metrics produced by the run, and
3. the post hoc validator-bank comparison against attested historical corpora not used in the optimization loop.

The paper does not claim blind reconstruction. Latin is in the optimization loop as the active target reference, and the run is therefore reported as a target-conditioned retrodictive search rather than as unsupervised ancestor recovery.

The present paper is best read as an instrument study rather than a contribution to historical-linguistic reconstruction. Language was chosen as the test domain because attested historical Romance corpora are available as held-out validators and because the author has a research interest in computational historical linguistics; the engine and the structural scoring path are likely agnostic to domain in their underlying structure and could in principle be applied to other corpus-statistical comparison tasks; the family alignment component, however, is specific to linguistic data. The contribution made here is therefore methodological: it documents a target-conditioned search over a specific four-dimensional feature space, the empirical behavior of that search against a held-out validator bank, and the metric-space properties that emerge under expansion to a richer feature space. The historical-linguistic content of those findings is a consequence of the chosen input data and active reference set, not a claim about Romance language change. The paper does not model historical language change in the wild: the transmission, contact, and innovation processes by which languages actually transform over time. The engine operates on corpus-level statistical features and is silent on those processes.

2 Materials and Methods

2.1 Corpus roles

The run reported here uses four corpus roles: source, target, held-out validators, and controls. Table 1 summarizes the role assignment used in the project-level methodology.

Table 1: Corpus roles used in the active methodology.

Role	Corpus set
Source corpus	Modern French
In-loop target	Classical Latin
Held-out historical validators	Old French, Middle French, Anglo-Norman, other langue d'oïl varieties, Old Spanish / early Iberian, Old Occitan
Null and control corpora	Markov noise, Sumerian, withheld Portuguese

The source corpus for the paper run was the processed modern French corpus in `data/processed/romance/french_tokens.json`. Latin served as the active reference for structural and form scoring. The validator bank was not used in the search objective and was only applied after the run completed. Sumerian was included in the control bank as a structured non-Indo-European control, intended to distinguish engine-driven structural tracking from the Markov-noise floor.

2.2 Statistical representation

Languages are represented as statistical fingerprints over tokenized corpora. The active representation combines the following components:

1. token co-occurrence structure,
2. positional token distributions,
3. word bigram and trigram profiles,
4. type-token statistics,
5. character bigram and trigram profiles, and
6. suffix profiles.

The structural side of the scoring path uses corpus-level structural vectors derived from these token-distribution features. The form side uses character-level profiles and suffix profiles. Candidate scoring in the engine therefore tracks at least three quantities throughout the run:

- Latin structural score,
- Latin form score,
- family alignment / coherence diagnostics.

Structural score. Each corpus state is projected onto a four-dimensional feature vector: type–token ratio, top-100 bigram coverage, top-100 trigram coverage, and $\log(1+\overline{L})$ where $\overline{L}$ is the mean sequence length. The structural score used as the in-loop search reward is the negative Euclidean distance between the candidate's and the target's vectors restricted to the first three features, scaled by 5.0:

$$S_{\text{struct}} = -5.0 \cdot \left\| \mathbf{v}_{\text{cand}}^{[1:3]} - \mathbf{v}_{\text{Latin}}^{[1:3]} \right\|_2 . \tag{1}$$

The fourth feature, $\log(1+\overline{L})$, is excluded from the reward because the candidate-generation path preserves the source sentence-length distribution and cannot optimize that dimension directly.

Zero indicates an exact three-feature match; the score is bounded above by zero. The full four-dimensional structural vector is retained for the post hoc validator-bank comparison described in Section 2.6, which uses a different distance function on a different reference set.

Form score. The form score combines three character-level cosine similarities against the target:

$$S_{\text{form}} = 0.40 \cdot \cos(\mathbf{bg}_{\text{cand}}, \mathbf{bg}_{\text{target}}) + 0.40 \cdot \cos(\mathbf{tg}_{\text{cand}}, \mathbf{tg}_{\text{target}}) + 0.20 \cdot \cos(\mathbf{sf}_{\text{cand}}, \mathbf{sf}_{\text{target}})$$

where **bg** is the top-1,500 character bigram profile, **tg** is the top-2,500 character trigram profile, and **sf** is the top-800 three-character suffix profile.

Family alignment score. The family alignment score applies the Hungarian assignment algorithm [5] to match suffix and prefix families between the candidate and the target, normalised to $[0, 1]$.

2.3 Search configuration

The run was executed through the *v5* long-run driver. The active engine inherits the *v4* mutation-and-acceptance controller and adds the *v5* configuration surface. For the paper run, the search used the configuration shown in Table 2.

Table 2: Frozen configuration for the French → Latin paper run.

Parameter	Value
Source language	French
Target language	Latin
Candidate count	16
Block proposals	1000
Number of sequences	800
Minimum improvement	0.0001
Structural target	0.0
Form target	1.0
Family target	1.0
Incremental scoring	on
Fortran cosine	on
Fortran batch	on
Semantic transparency	off
Transparency weight	0.0
Culture bombs	off
Validator snapshots during run	off
Live event mode	all
Requested launch seed	45
Audited engine seed	42

Candidate proposals were evaluated under the active *v5* weighted objective. The total score for a candidate state combines the Latin structural and form scores defined in §2.2, a coherence margin against the Markov-noise floor, the candidate's mutation cost, and a reward bonus that rewards strict improvement along the structural, form, suffix, and character-trigram axes:

$$S_{\text{total}} = S_{\text{struct}} + w_f\, S_{\text{form}} + w_c\, M_{\text{coh}} - w_m\, C_{\text{mut}} + R_{\text{bonus}} + R_{\text{relief}}, \tag{2}$$

with reward and relief terms

$$\begin{aligned} R_{\text{bonus}} &= w_{\Delta s}\,[\Delta s]_+ + w_{\Delta f}\,[\Delta f]_+ + w_{\Delta\sigma}\,[\Delta\sigma]_+ + w_{\Delta\tau}\,[\Delta\tau]_+ + b_j\,\mathbf{1}_J, \\ R_{\text{relief}} &= w_r\, w_m\, C_{\text{mut}}\,\mathbf{1}_J, \end{aligned}$$

where $[x]_+ = \max(0, x)$; Δs, Δf, $\Delta\sigma$, and $\Delta\tau$ are the changes in Latin structural score, Latin form score, character suffix cosine, and character trigram cosine between the candidate and the current accepted state; and $\mathbf{1}_J$ is the indicator for the joint-gain condition $(\Delta s > 0) \wedge (\Delta f > 0) \wedge (\Delta c \geq -\delta_c)$, with Δc the coherence-margin gain and δ_c a fixed tolerance configured separately. The numerical values used for the paper run were:

- $w_f = 0.75$ (form weight),
- $w_c = 0.05$ (coherence weight),
- $w_m = 0.005$ (mutation cost weight),
- $w_{\Delta s} = 8.0$ (reward struct gain weight),
- $w_{\Delta f} = 4.0$ (reward form gain weight),
- $w_{\Delta\sigma} = 2.0$ (reward suffix gain weight),
- $w_{\Delta\tau} = 2.0$ (reward trigram gain weight),
- $b_j = 0.01$ (reward joint bonus),
- $w_r = 1.0$ (reward penalty relief).

Accepted-stage dense artifact generation remained in the standard Python path. The accelerated path moved hot-loop scoring work into incremental and Fortran-assisted components without changing the run's methodological objective.

2.4 Computational execution path

Within each proposal cycle, the engine generated a fixed batch of candidate mutations, scored those candidates under the active objective, accepted at most one candidate, and then advanced to the next proposal. The paper run used the following division of labor:

- Python owned mutation generation, acceptance decisions, block control, artifact writing, and manifest updates.
- Incremental scoring state maintained rolling counters so that unchanged portions of the corpus did not need to be rescored from scratch on every candidate.
- Fortran-backed paths were enabled for cosine-heavy structural computations and candidate-batch structural scoring.
- Dense accepted-stage artifact generation remained in the Python path and was only performed for committed stages rather than for every rejected candidate.

This split preserves a single methodological objective while moving the highest-frequency numeric work into the accelerated path.

2.5 Stopping rule

The run used indefinite total budget (`total_target_proposals = 0`) with plateau-based stopping. The plateau rule operated over a window of 10 completed blocks. For each of the three monitored axes

- Latin structural score,
- Latin form score,
- family alignment score,

the driver compared the best value observed in the most recent 10 blocks to the best value observed in all prior blocks. Plateau was declared if none of the three recent best values exceeded the earlier best by more than its threshold:

- structural epsilon = 0.001,
- form epsilon = 0.001,
- family epsilon = 0.001.

The run did not reach the joint target condition ($\texttt{struct} \geq \texttt{0.0}$, $\texttt{form} \geq \texttt{1.0}$, $\texttt{family} \geq \texttt{1.0}$). It halted by `plateau_hit`.

2.6 Held-out validator-bank analysis

After the run completed, one selected checkpoint per completed block was compared against every active attested validator corpus. The active validator bank for this analysis contained the six corpora listed in Table 3.

Table 3: Active attested validator bank used in the post hoc comparison.

Corpus	Branch	Date range	Region	Tokens
Old French	French	842–1270	France	80,049
Middle French	French	1450–1489	France	31,818
Anglo-Norman	French	1160–1200	England / Normandy	27,612
langue d'oïl	French	1150–1225	Northern France	33,873
Old Spanish	Spanish	1140–1270	Iberia	29,651
Old Occitan	Occitan	1200–1229	Occitania	70,826

For each block-validator pair, the analysis recorded:

- validator structural cosine,
- validator structural distance,
- validator form score,
- validator character bigram cosine,
- validator character trigram cosine,

- validator suffix cosine.

The validator-bank analysis operates on the full four-dimensional structural feature vector, including the fourth feature ($\log(1 + \overline{L})$) excluded from the in-loop reward (Section 2.2). The "validator structural distance" reported in Tables 6 and 7 below is the scaled Euclidean distance between candidate and validator 4D vectors, with per-feature normalization by the standard deviation of the real-language reference set. This is mathematically distinct from the in-loop reward, which operates in a three-feature subspace and uses an unnormalized Euclidean form. The two quantities live on different numerical scales: a near-zero in-loop reward score at the run endpoint and validator structural distances near 2 are not contradictory; they measure different geometric relationships between candidate and reference under different distance functions.

The validator-bank analysis emitted CSV, JSON, and chronology summaries. The current paper reports the chronology summary and final-block rankings.

3 Results

The results below describe the completed run under two complementary readings. The trajectory is first reported in chronological terms via the validator-bank nearest neighbors, then re-examined under a typological reading supported by the post hoc dimensional robustness check (Section 3.7). The validator-bank chronology is the natural descriptive level for the active medieval-French validator bank, but the typological reading is what the dimensional robustness check licenses and is the framing this paper adopts in its conclusion.

3.1 Run completion and endpoint

The paper run halted with manifest status `plateau_hit` after 614 blocks and 614,000 proposals. Average throughput over the full run was 4,533.74 proposals per hour, for a total wall-clock runtime of approximately 135.4 hours.

The final endpoint metrics were:

- Latin structural score: -0.00010808926278147055,
- Latin form score: 0.7611594796180725,
- family alignment score: 0.513866759690557.

Under the active structural metric, the final endpoint therefore achieves a near-zero Latin structural score.

3.2 Preserved checkpoints

The run preserved multiple endpoints of interest. Table 4 summarizes the three checkpoints carried forward for analysis.

The best-form checkpoint occurred at block 338. The final structural endpoint was preserved separately from the final plateau endpoint because the near-zero structural minimum was already present in block 613. The best-form and best-structure checkpoints therefore did not coincide in the same block.

Table 5 shows representative corpus excerpts at each preserved checkpoint. The excerpts are taken verbatim from the block-level preview files and illustrate the progression of the structural

Table 4: Primary preserved checkpoints from the paper run.

Checkpoint	Block	Structural score	Form score	Family alignment
Seed-adjacent early state	0001	−1.3435869472793973	0.7181036472320557	0.4392205108144682
Best form checkpoint	0338	−0.8288512723323529	0.8632466197013855	0.6087353051690383
Best structural checkpoint	0613	−0.00010808926278147055	0.7607623934745789	0.5140366151364617
Final plateau endpoint	0614	−0.00010808926278147055	0.7611594796180725	0.5138667596905570

transformation: block 0001 remains recognizably modern French with occasional Latin-style suffix intrusions, block 0338 shows a mid-transformation state with mixed French-Latin morphology, and blocks 0613 and 0614 show heavy suffix substitution toward Latin-style endings.

Table 5: Representative sample sequences from each preserved checkpoint, taken verbatim from the block-level corpus preview files.

Block	Sample sequences
0001	*porta il du brésil porta il des routes* *aes henan partage ses frontières avec six autres provinces*
0338	*flash riuit mi flash série télévisbus américaines* *qui sere officitur*
0613	*pris nerius rentum que pris nerius que* *artia situm mere quam sis que*
0614	*situm télévitum* *photogratum*

3.3 Validator-bank chronology

The chronology below treats each block's nearest validator as a stand-in for "what historical stage of French the corpus resembles at that block." This is a chronological reading. The typological reading of the same trajectory is presented in Section 3.7.

The validator-bank chronology summary for the complete run reported the following nearest-neighbor paths:

- structural path: Middle French → Old Spanish → Middle French → Old Spanish → Middle French → Old Spanish → Middle French → Old Spanish → Old Occitan,
- form path: Middle French → langue d'oïl → Middle French → langue d'oïl → Middle French.

The first structural nearest-neighbor change points recorded in the chronology summary were:

- Middle French first structural match: block 0001,
- Old Spanish first structural match: block 0029,
- Old Occitan first structural match: block 0379.

The first form nearest-neighbor change points recorded in the chronology summary were:

- Middle French first form match: block 0001,
- langue d'oïl first form match: block 0365.

3.4 Final-block validator rankings

At the final block, the top structural and form matches were those shown in Tables 6 and 7.

Table 6: Top structural matches at the final block (block 0614).

Validator	Date range	Structural distance
Old Occitan	1200–1229	1.9953
Old French	842–1270	2.0242
langue d'oïl	1150–1225	2.3768

Table 7: Top form matches at the final block (block 0614).

Validator	Date range	Validator form score
Middle French	1450–1489	0.4213
langue d'oïl	1150–1225	0.4125
Anglo-Norman	1160–1200	0.4000

The chronology summary also records the best structural block achieved against each validator. The minimum validator structural distances observed anywhere in the run were:

- Old French: 1.9405 at block 0465,
- Middle French: 2.0302 at block 0384,
- Anglo-Norman: 2.2559 at block 0434,
- langue d'oïl: 2.0476 at block 0405,
- Old Spanish: 1.9790 at block 0383,
- Old Occitan: 1.7622 at block 0424.

3.5 Validator-bank summary pattern

Under the active validator bank, the trajectory does not reproduce a simple reverse French chronology. The structural nearest-neighbor oscillates between Middle French and Old Spanish across the first 379 blocks before transitioning to Old Occitan, which remains nearest through the plateau. The form nearest- neighbor stays in the French / langue d'oïl region throughout the run, with langue d'oïl varieties becoming relatively more frequent near the end as the structural trajectory moves away from Modern French. The final block is structurally nearest to Old Occitan and form-wise nearest to Middle French.

These observations are reported here without mechanistic interpretation. Any proposed explanation of the trajectory would need to satisfy three constraints simultaneously. First, it must be grounded in what the four structural features actually measure (type-token ratio, top-100 bigram and trigram coverage, and log mean sequence length); explanations that invoke phonological or morphological historical mechanisms only qualify if those mechanisms demonstrably manifest in those four quantities. Second, it must account for the Portuguese control result: the minimum structural distance achieved against any medieval validator in this run is 1.7622 (Old Occitan, block 0424), while withheld Modern Portuguese achieves 0.9730 (block 0474), roughly twice as close as the nearest medieval language; an explanation of validator proximity that does not address this gap treats the stronger signal as invisible. Third, it must account for both paths jointly: the run ends structurally nearest to Old Occitan while remaining form-nearest to Middle French throughout, and these two trajectories diverge steadily in the second half of the run. No explanation known to the author at the time of the original validator-bank pass satisfied all three constraints simultaneously. A candidate interpretation, supported by a post hoc dimensional expansion of the structural feature space, is reported in Section 3.7 below.

3.6 Control-bank summary pattern

A post hoc control-bank pass compared all 614 completed block endpoints against three non-target controls: Markov noise, Sumerian, and withheld Portuguese. Under that comparison, the nearest structural control followed the path Sumerian → Portuguese, while the nearest form control was Portuguese throughout. Markov noise never became the nearest control on either axis.

At the final block, the control rankings were:

- structural distance: Portuguese 1.1293, Sumerian 2.5902, Markov 3.5248;
- control-bank form score: Portuguese 0.3553, Sumerian 0.1045, Markov 0.0.

The first structural Portuguese match occurred at block 0300. The best structural distances observed anywhere in the run were 0.9730 against Portuguese (block 0474), 2.3184 against Sumerian (block 0408), and 3.5248 against Markov noise (block 0613).

Figure 1 shows the per-block trajectory of structural distance to each of the three controls across the full run. All three control distances decrease over the run. Portuguese and Sumerian track each other closely through the first 300 blocks, after which Portuguese diverges sharply downward to a minimum near block 470, while Sumerian declines more gradually to a minimum near block 408 and then plateaus. Markov noise decreases least and remains well separated from the other two controls throughout.

3.7 Dimensional robustness of the structural fingerprint

The structural metric used in this paper operates in a four-dimensional feature space. A natural question is whether the nearest-neighbor rankings reported above are properties of the corpora or artifacts of compressing inter-corpus structure into four dimensions. A post hoc dimensional robustness check was performed against this concern; the full report is stored at `docs/reports/2026-05-19_dimensional_robustness_check.md` and the numeric outputs at `data/validation/2026-05-19_dimensional_robustness/`.

The check expanded the structural feature set to fifteen dimensions (the original four plus eleven additional features: top-500 word bigram and trigram coverage, top-1000 unigram coverage, sequence-length variance, word-length mean and variance, hapax legomena ratio, Zipf slope over ranks 10–1000, word bigram and trigram entropy, and suffix-profile entropy), z-scored across the scored corpora, and recomputed cosine distance to Latin under 4D, 10D, and 15D for the validator

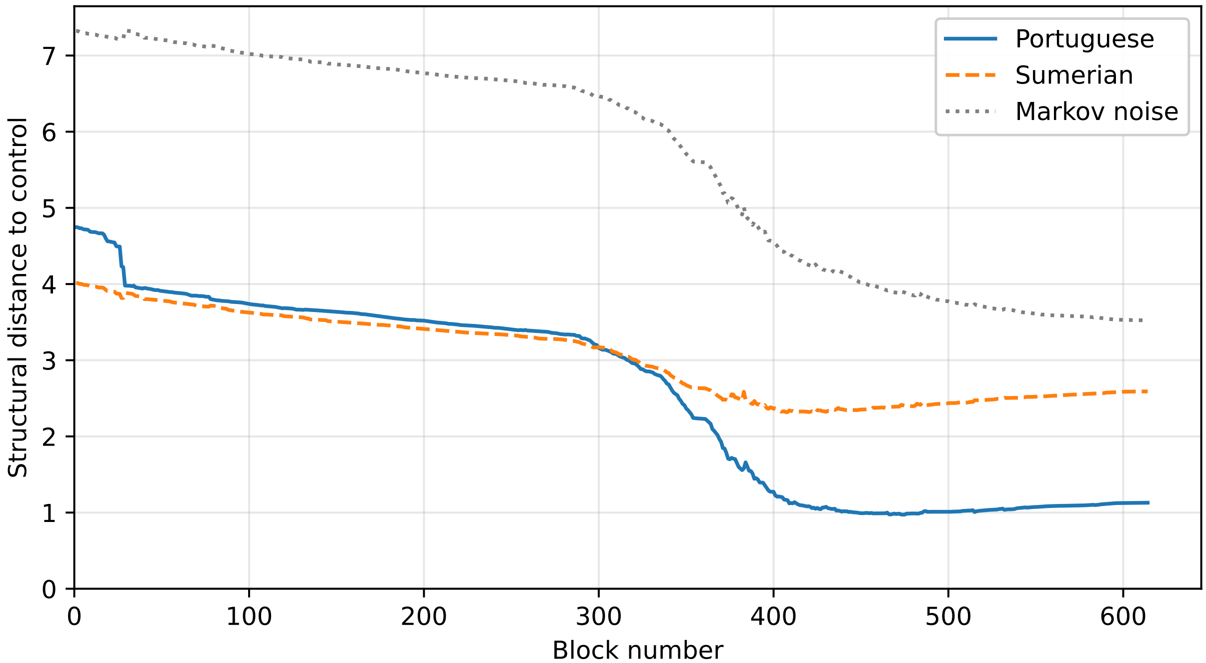

Figure 1: Per-block structural distance from the synthetic corpus state to each of the three controls (Portuguese, Sumerian, Markov noise) across the 614-block paper run. Portuguese reaches its minimum near block 470 at 0.9730. Sumerian reaches its minimum near block 408 at 2.3184. Markov noise reaches its minimum at the final block at 3.5248. Portuguese is never above the Sumerian curve in the second half of the run, but Sumerian remains substantially closer to the synthetic state than Markov noise throughout.

bank, the control bank, and the v5 paper-run endpoint. The 4D scaled-Euclidean distance from Modern Portuguese to block 0474 was reproduced exactly (0.9729932421288967), confirming that the expanded analysis uses the same loading and feature path as the production scoring code. Table 8 reports the cosine distances under 4D and 15D, sorted by 15D rank.

Principal-components analysis on the z-scored 15-feature matrix places 95% of inter-corpus variance in the first five components, with the fifth component carrying 3.9% and loading on top-1000 unigram coverage and word-length mean, features that the original 4D space did not include. The intrinsic dimensionality of inter-corpus structure for the corpora scored here is therefore approximately five, not four.

The Sumerian result. The single most consequential observation from the dimensional expansion is the position of Sumerian, a corpus originally selected as a control because it was presumed unlikely to cluster near Latin under any meaningful metric. This was not a result the run was designed to produce, and the interpretation offered below is post hoc. In retrospect, it may not have been the least rhetorically hazardous control language available; Japanese or Quechua would have made the same typological point. Under 4D the metric placed it at rank 14, last among the ten corpora; under 15D it rises to rank 4, inside the medieval Romance cluster on structural proximity to Latin. Sumerian is non-Indo-European, agglutinative, and historically unrelated to the Romance line; no genealogical or contact-based account places Sumerian closer to Latin than

Table 8: Cosine distance to Latin under 4D and 15D z-scored structural feature spaces, sorted by 15D rank. The 4D space corresponds to the production structural metric used in this paper; the 15D space adds eleven Tier-1 and Tier-2 features documented in the robustness report.

Corpus	4D dist	15D dist	4D rank	15D rank
Modern Portuguese (withheld)	0.813	**0.334**	7	**1**
Modern French (source)	0.812	0.577	6	2
Old Occitan	0.624	0.799	4	3
Sumerian	1.539	0.800	14	4
Old French	0.340	0.904	1	5
Old Spanish	1.001	0.954	10	6
langue d'oïl	0.725	1.048	5	7
Markov noise	0.553	1.080	2	8
Middle French	1.062	1.091	11	9
Anglo-Norman	0.575	1.213	3	10

Old French or langue d'oïl under any plausible historical metric. Its rank-4 position under the richer feature space is therefore consistent with reading the structural metric as a detector of typological proximity (morphological richness, suffix-profile entropy, inflectional vocabulary breadth) rather than genetic relationship or historical contact. We treat this result as the strongest available evidence against a purely genealogical reading of the metric. The Portuguese-vs-medieval-validator comparison reported below is consistent with that reading; the Sumerian rank is what rules out a purely genealogical alternative.

The expansion produces two further consistent observations. First, the Portuguese-vs-medieval-validator comparison strengthens rather than weakens under expansion: the 4D space muted the Portuguese-Latin proximity, and the eleven new features overwhelmingly pull Portuguese further toward Latin. Second, the block-state ordering of v5 paper-run checkpoints is preserved across all three dimensional settings: the search did not climb an artifact of the 4D compression.

Reading the structural metric as a typological proximity detector rather than a chronological one satisfies the three constraints articulated above. The feature-space constraint is satisfied because typological proximity is a plausible reading of what those four features collectively measure. The Portuguese constraint is satisfied because Modern Portuguese has retained more Latin inflectional morphology than Modern French has, and that retention manifests in TTR, unigram coverage, word-length, and suffix-entropy axes. The path-bifurcation constraint is satisfied because the structural and form axes measure different properties (corpus-level morphological similarity versus character-level surface similarity), which need not be co-monotonic on a target-conditioned trajectory starting in a less-inflected modern descendant. The interpretation is offered as the most parsimonious available reading of the present data, not as a mechanistic claim about how Romance languages stand to Latin.

4 Limitations

The present report is subject to the following constraints:

1. The run is target-conditioned, not blind. Latin is in the active optimization loop.
2. The validator bank currently contains six attested medieval Romance corpora. Token counts

range from approximately 27,600 (Anglo-Norman) to 80,000 (Old French).

3. The Old Spanish validator corpus is dominated by a single author and genre: Gonzalo de Berceo's *Milagros de Nuestra Señora* (26 miracles plus prologue) plus the *Poema de Mio Cid* fragment and related texts. The resulting fingerprint may reflect Berceo's individual style as much as the broader Old Spanish register.

4. The control bank currently contains three controls only: Markov noise, Sumerian, and withheld Portuguese.

5. The historical seed discrepancy has been resolved by replay audit, but the original launch path for the paper run did not preserve a self-consistent requested-seed versus engine-seed record at run time.

6. The paper reports one source-target run only: French $\rightarrow$ Latin.

7. A three-seed early-window replication probe (seeds 42, 43, and 45, 1,000 proposals each) found that all three seeds produced coherent, same-direction progress over the first block, with final structural scores in the range -1.3614 to -1.3062 and form scores in the range 0.6928 to 0.7181. Seed 42 reproduced the historical paper-run block 0001 endpoint exactly. This probe supports the bounded claim that early-window behavior is not specific to a single seed; it does not establish whether the full long-horizon phase structure or the final plateau endpoint would replicate across seeds.

8. The structural metric used during the paper run is four-dimensional. A post hoc dimensional robustness check (Section 3.7) finds the intrinsic dimensionality of inter-corpus variance to be approximately five, so the 4D space is slightly undersized for the full set of structural comparisons. The Portuguese-vs-medieval-validator comparison survives expansion to fifteen dimensions, but quantitative comparisons among the medieval validators themselves should be interpreted with caution, since some rank order changes under expansion.

9. The family alignment score has a structural ceiling below 1.0 that depends on the target language. A post hoc self-similarity calibration finds the matched-slice ceiling for Latin to be 0.9583: six of Latin's top-12 short-token families (*in*, *et*, *ut*, *ad*, *me*, *si*) are 2-character tokens shorter than the suffix length used by the Hungarian alignment cost function, producing empty suffix profiles and a fixed self-cost of 0.0417. The v5 endpoint family alignment score of 0.5139 therefore represents approximately 54% of the achievable Latin ceiling, not 51% of nominal. Per-corpus ceilings vary between 0.9375 and 0.9722 depending on how many of each corpus's top-12 short-token families are 2-character entries; details are documented in `docs/reports/2026-05-19_latin_self_similarity_calibration.md`.

10. Adversarial baselines beyond the Markov-noise null (feature ablations dropping individual structural components, shuffled feature-weight controls, synthetic-corpus controls preserving Zipf and length distributions, and randomized target-feature trajectories) were not run for this paper. The Markov, Sumerian, and Portuguese controls test that the optimizer finds non-trivial structure and that the structure is not lineage-tracking, but they do not test which individual features drive the structural metric. These ablations are a natural next experiment.

5 Conclusion

The structural metric described in this paper appears to function as a typological proximity detector rather than a chronological one. A post hoc expansion of the structural feature space to fifteen dimensions (Section 3.7) preserved the Portuguese-vs-medieval-validator comparison and additionally placed Sumerian, a non-Indo-European agglutinative language with no historical relationship to the Romance line, at fourth-rank structural proximity to Latin. The instrument behavior reported here is consistent with that reading and does not support mechanistic claims about Romance historical linguistics.

The run that produced this evidence was a complete French → Latin retrodictive search under a frozen *v5* configuration. The run halted by plateau after 614,000 proposals and reached a near-zero Latin structural score under the active metric. The best form checkpoint occurred earlier than the final plateau endpoint. Post hoc validator-bank comparison reported a structural nearest-neighbor path that oscillated between Middle French and Old Spanish for the first 379 blocks before settling on Old Occitan through the plateau endpoint, and a form nearest-neighbor path concentrated in Middle French and other langue d'oïl material, with the final block structurally nearest to Old Occitan and form-wise nearest to Middle French.

Post hoc control-bank comparison reported a structural control path of Sumerian → Portuguese and a form control path of Portuguese, while Markov noise never became the nearest control on either axis. Withheld Modern Portuguese reached a structural distance roughly twice as close to the final block as the nearest medieval Romance validator, a gap that any historical-reconstruction reading of the validator-bank chronology would need to address.

Reproducibility Note

The run manifest, preserved corpora, previews, block summaries, and validator-bank outputs are stored in the repository. The paper-facing run terminated by a native plateau rule rather than by manual truncation. A post hoc replay audit resolved the historical seed mismatch for this run: the manifest launch field records requested seed 45, while exact replay of block 0001 identifies engine seed 42 as the seed actually used by the search. The full audit is documented in `docs/reports/2026-04-24_v5_paper_run_seed_audit.md`. The validator-bank comparison files cited in this paper are stored under `data/validation/french_v5_fortran_c16_seed45_paper_run_validator_bank_vs_validator_bank.*`. The control-bank comparison files used for Figure 1 and Section 3.7 are stored under `data/validation/french_v5_fortran_c16_seed45_paper_run_vs_control_bank.*`. The post hoc dimensional robustness check is documented in `docs/reports/2026-05-19_dimensional_robustness_check.md`, with numeric outputs under `data/validation/2026-05-19_dimensional_robustness/`. The Latin self-similarity calibration is documented in `docs/reports/2026-05-19_latin_self_similarity_calibration.md`, with numeric outputs under `data/validation/2026-05-19_self_similarity/`.

Data Availability

All source code, manuscript source, and lightweight analysis artifacts supporting this study are publicly available at `https://github.com/trafaelosborn/project-rbt` and archived at Zenodo under DOI 10.5281/zenodo.20330397. The Zenodo deposit additionally includes the Latin token reference file (`data/sequestered/latin/latin_tokens.json`), which exceeds GitHub's per-file size limit and is therefore distributed exclusively through Zenodo. No restrictions apply to access or reuse beyond the terms of the MIT license under which the repository is released.

Competing Interests

The author declares no competing interests.

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Author Contributions

T.O. designed the study, performed the analysis, and wrote the manuscript.

Use of AI Tools

The author designed the methodology, experiments, and overall research program. Claude Code (Anthropic) drafted most of the implementation, including substantial portions of the engine (`src/retrodiction/*.py`, `src/validation/*.py`), the post hoc dimensional robustness check script (`tmp_dimcheck/dim_robustness.py`), and the Latin self-similarity calibration script (`tmp_self_similarity/calibration.py`). Manuscript writing, restructuring, and editing (including drafting of Section 3.7, the constraints paragraph in §3.5, the v5 objective equation in §2.3, the Conclusion paragraph on the typological-detector reading, the related-work expansion in §1, and the §2.6 metric-scale clarification) were assisted by Claude (Anthropic) operating in the Octave document-chat workstation. All AI contributions followed prompts and specifications written by the author; the author retains full responsibility for all design choices, implementation correctness, analysis, and conclusions.

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